Wave Energy and Shoreline Protection

Lecture notes, Fall 2026

1. Energy in a wave

For a linear wave of height H, the energy per unit area of sea surface is

$$E = \frac{1}{8}\rho g H^2$$

where ρ is the density of seawater and g is gravitational acceleration. Because energy scales with the square of height, halving the wave height removes three quarters of its energy.

2. Dissipation through vegetation

As a wave crosses a marsh, drag on the stems removes energy. Over short distances the height decays approximately exponentially:

$$H(x) = H_0\, e^{-k x}, \qquad k = \frac{C_D\, b\, N}{3\pi}\,\frac{H_0\, k_w}{h}$$

Here C_D is the drag coefficient, b the stem diameter and N the stem density.

3. Typical values

Shoreline	k (m⁻¹)	Height left after 20 m
Bulkhead	0.004	92%
Oyster sill	0.018	70%
Marsh and sill	0.031	54%

Exercise. Using the table, how much wave energy remains 20 m behind a marsh and sill?